Algorithmic Trading

Python Suite: Post-Trade Analysis

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HW 06

Assignment:

- Develop Python TCA Post Trade functions to calculate the following:
  - Implementation Shortfall
  - Paper Return
  - Portfolio Return
  - Arrival Cost
  - VWAP Slippage
  - Benchmark Cost
  - RPM (Relative Performance Measure)
  - Value-Add
- Answer all questions using your Python Code.
- See "TCA_Important Formula.pdf" for the TCA formulas.
- You must submit your Python functions to BB to receive credit for this assignment.

Q1) Implementation Shortfall

Calculate Paper Return, Portfolio Return, Implementation Shortfall, and all Implementation Shortfall components for the following order: Side = Buy, S = 200,000, X= 180,000, Pd = $60.00, P0= $60.05, Pn = $60.65, Pavg= $60.45, Commission = $0.01/share.

- Paper Return= 130,000; Portfolio Return = 34,200; Paper Return – Actual Return = 95,800;
- Implementation Shortfall = 95,800; Delay Cost = 10,000; Execution Cost = 72,000; Opportunity Cost = 12,000; Fixed Cost = 1,800.

Q2) Arrival Cost

A manager bought 100,000 shares of stock RLK at an average price of Pavg=$25.30. The stock opened at a price of Open=$25.05. The arrival price when the order was released to the market was P0=$25.15. The VWAP price over the trading horizon was VWAP=$25.22. The closing price on the day was Close=$25.40. What is the Arrival Cost of the order?

- Arrival Cost = 59.6421

Q3) VWAP Slippage

The average price for a sell order was Pavg=$55.37. The stock opened at a price of $55.50. The arrival price when the order was released to the market was $55.45. The VWAP price over the trading horizon was $55.35. The closing price on the day was Close=$55.30. What is the VWAP Slippage for this order?

- VWAP Slippage = -3.6133

Q4) Benchmark

A portfolio manager bought 50,000 shares of RLK at an average price of $150.65. RLK opened at Open=$150.50. The arrival price when the order was released to the market was P0=$150.55. The VWAP price over the trading horizon was VWAP=$150.62. The closing price on the day was Close=$150.95. What is the performance compared to the closing price on the day? (Hint: the Closing Price is the reference price).

- Close Benchmark Cost = -19.8741

Q5) Value-Add

A portfolio manager bought 250,000 shares of RLK at an average price of Pavg=135.35. The arrival price of the order was P0=$134.90. The estimated market impact and timing risk for this order was MI=30bp and TR=40bp. Calculate the Value-Add and Z-Score for this order?

- Value-Add = -3.3580, ZScore = -0.0839

Q6) Value-Add

A trader attempted to buy 100,000 shares of stock RLK but was only able to execute 60,000 shares. The arrival price (price when the order was entered into the market) was P0=$51.00 and the average execution price was Pavg=$51.22.

- The estimated market impact and timing risk for 100,000 shares of RLK was MI=50bp and TR=100bp.
- The estimated market impact and timing risk for 60,000 shares of RLK was MI=35bp and TR=65bp.

Calculate the Value-Add and Z-Score for this order?

- Value-Add = -8.1372, ZScore = -0.1251

Q7) RPM

Using the Trade Information table, calculate the RPM for the following orders:

a) Buy order for 6,500 shares with an average execution price of Pavg=$25.50. RPM-Buy = 44%

b) Sell order for 10,000 shares with an average execution price of Pavg=$24.95. RPM-Sell = 33%

Trade Information	
Price	**Volume**
24.50	5,000
26.50	2,500
27.50	9,000
25.25	4,000
25.50	5,000
26.00	3,000
25.00	5,000
27.10	5,000
24.00	9,000
24.75	2,500

Q8) RPM

Using the Trade Information table, calculate the RPM for the following orders:

a) Buy order for 25,000 shares with an average execution price of Pavg=$65.25. RPM-Buy = 42.4%

b) Sell order for 35,000 shares with an average execution price of Pavg=$65.33. RPM-Buy = 79.2%

Trade Information	
Price	**Volume**
65.15	25,000
65.30	85,000
65.35	35,000
65.25	100,000
65.45	40,000
65.00	15,000
65.40	50,000
65.50	5,000
65.10	80,000
65.20	65,000
65.05	125,000